Structured Pruning and Post-Training Quantization for Lightweight Vision Transformers on Edge Devices

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***Abstract*—Vision Transformers (ViTs) have achieved remarkable performance across various computer vision tasks, but their computational demands remain a barrier to edge deployment. This paper systematically investigates the compounding effect of structured pruning and post-training quantization (PTQ) on lightweight ViT variants. We evaluate DeiT-Tiny under various pruning ratios (25–50%) combined with INT8 dynamic quantization. Our results show that combining 50% head+MLP pruning with PTQ reduces model size from 21.9MB to 3.4MB (84.4% reduction) while cutting FLOPs by 54%, demonstrating viability for resource-constrained edge deployment.**

I. Introduction

Vision Transformers (ViTs) [1] have become the dominant architecture for computer vision, but their large parameter count and high computational cost hinder deployment on edge devices. While quantization [2] and pruning [3] are well-studied individually, their compounding effect on tiny ViTs remains underexplored. We address this gap.

II. Methodology

We apply structured pruning to DeiT-Tiny [4]:

- **Head pruning:** Remove entire attention heads by truncating the QKV projection matrix and updating `num_heads`.
- **MLP pruning:** Reduce the intermediate dimension of feed-forward networks by slicing `fc1` and `fc2` weights.
- **Dynamic PTQ:** Apply INT8 dynamic quantization (`torch.ao.quantization.quantize_dynamic`) on CPU after pruning.

Ten configurations are benchmarked covering head ratios 1.0, 0.75, 0.5 and MLP ratios 1.0, 0.75, 0.5 with and without PTQ.

III. Results

Table I summarizes the key findings.

TABLE I
DeiT-Tiny pruning + PTQ results

Config	Params(M)	Size(MB)	FLOPs(M)	TP(img/s)
Baseline	5.72	21.87	1074.85	194
H0.75_M0.75	4.24	20.27	784.36	279
H0.50_M0.50	2.76	18.58	493.87	178
PTQ-only	5.72	6.26	1074.85	102*
H0.75_M0.50_Q	3.35	3.98	610.07	145*
H0.50_M0.50_Q	2.76	3.41	493.87	162*

*PTQ latency on CPU; others on CUDA. TP = Throughput

The combination of 50% head+MLP pruning with PTQ achieves 84.4% model size reduction (21.9MB $\rightarrow$ 3.4MB) and 54.1% FLOPs reduction, making DeiT-Tiny feasible for deployment in sub-4MB flash budgets typical of MCU-class devices.

IV. Conclusion

Structured pruning and post-training quantization exhibit compounding benefits for tiny ViTs. The combined pipeline reduces model size by 5–6$\times$ while halving computational cost, all without fine-tuning or retraining. Future work will extend this to additional architectures and evaluate on physical edge hardware.

References

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